Day 11 – Kali Linux Setup (Theory Assignment)

Recommended file name: `StudentName_Day11_KaliSetup.pdf`

Below are answers to all 20 questions in simple, student-friendly language, written in original words.

THEORY – Answers

1. What is Kali Linux?

Kali Linux is a Debian-based Linux operating system designed mainly for cybersecurity, ethical hacking, penetration testing, and digital forensics. It comes with many pre-installed security tools.

2. Why is Kali Linux used in cybersecurity?

Kali Linux is used in cybersecurity to identify security weaknesses, test network security, analyze systems, detect vulnerabilities, and perform ethical hacking.

3. What is a virtual machine?

A virtual machine (VM) is a software-based computer that runs inside a physical computer. It has its own operating system, memory, storage, and virtual hardware.

4. What is a hypervisor?

A hypervisor is software that creates and manages virtual machines. It allows multiple operating systems to run on a single physical computer.

5. What is VirtualBox?

VirtualBox is a free and open-source virtualization software developed by Oracle. It allows users to install and run different operating systems as virtual machines on their computers.

6. Explain Host OS and Guest OS.

- Host OS: The main operating system installed directly on the physical computer.
- Guest OS: The operating system installed and running inside a virtual machine.

For example, Windows can be the Host OS, while Kali Linux can be the Guest OS.

7. Why is using Kali inside a VM useful for beginners?

Using Kali Linux inside a virtual machine is useful for beginners because it provides a safe learning environment. Students can practice cybersecurity without directly affecting their main operating system.

8. What is a VM snapshot?

A VM snapshot is a saved state of a virtual machine at a particular moment. It stores information about the VM so that the user can return to that previous state later.

9. Why should students create snapshots?

Students should create snapshots before experimenting with new tools or changing system settings. If something goes wrong, they can restore the virtual machine to its earlier working condition.

10. What does `whoami` display?

The `whoami` command displays the username of the currently logged-in user in the Linux system.

11. What does `hostname` display?

The `hostname` command displays the name assigned to the computer or system on the network.

12. What does `uname -a` display?

The `uname -a` command displays complete system information, including the operating system kernel name, hostname, kernel version, processor architecture, and other details.

13. What does `/etc/os-release` contain?

The `/etc/os-release` file contains information about the installed Linux distribution, such as its name, version, ID, and other operating system details.

14. What does `ip addr show` display?

The `ip addr show` command displays information about network interfaces, including their IP addresses, interface names, and network status.

15. What does `ip route show` display?

The `ip route show` command displays the system's routing table. It shows how network traffic is directed to different networks and the default gateway.

16. What is sudo?

`sudo` stands for "Superuser Do." It allows an authorized user to execute commands with administrator or elevated privileges.

17. Why should elevated privileges be used carefully?

Elevated privileges should be used carefully because administrator-level commands can modify important system files, change security settings, delete data, or damage the operating system.

18. What does `sudo apt update` do?

The `sudo apt update` command refreshes the package information from configured software repositories. It helps the system identify the latest available software versions and updates.

19. Why should Kali Linux tools only be used on authorized systems?

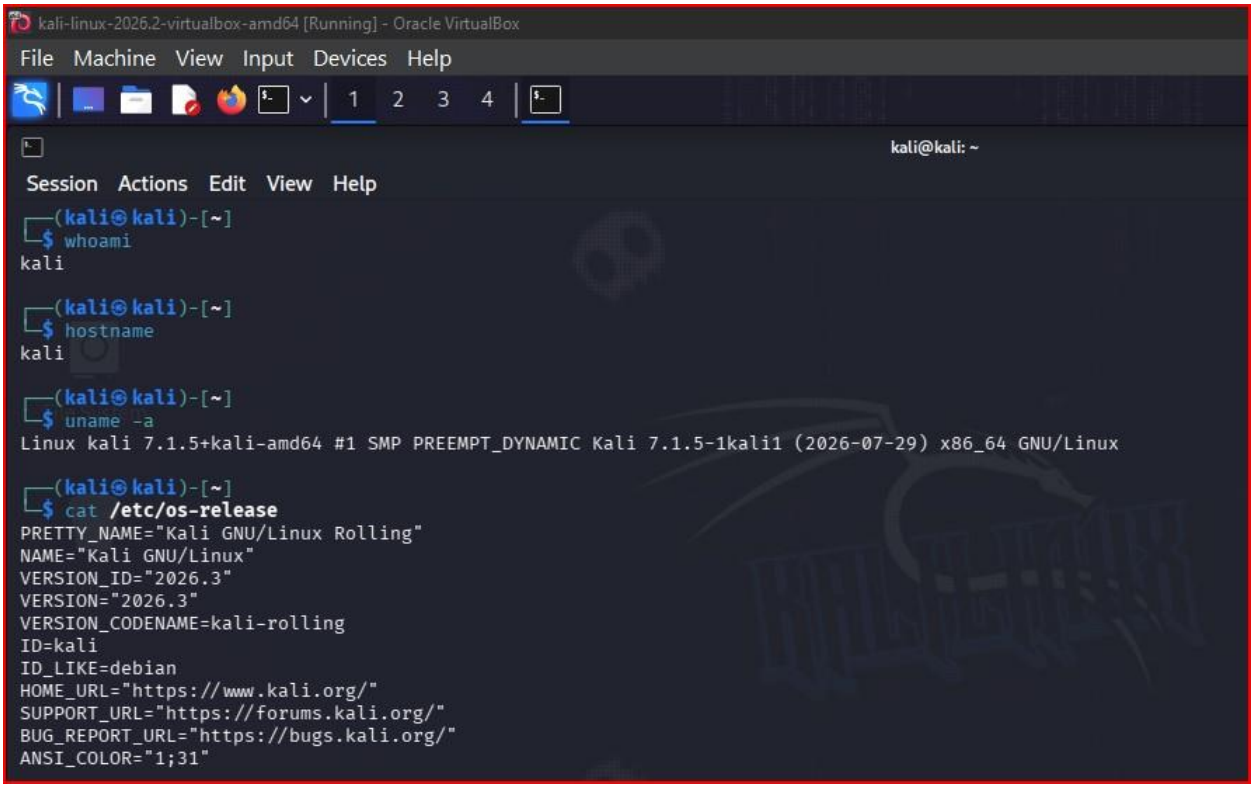
Kali Linux tools should only be used on authorized systems because using them without permission can violate privacy, damage systems, and break cybersecurity laws. Permission ensures that security testing is performed legally and responsibly.

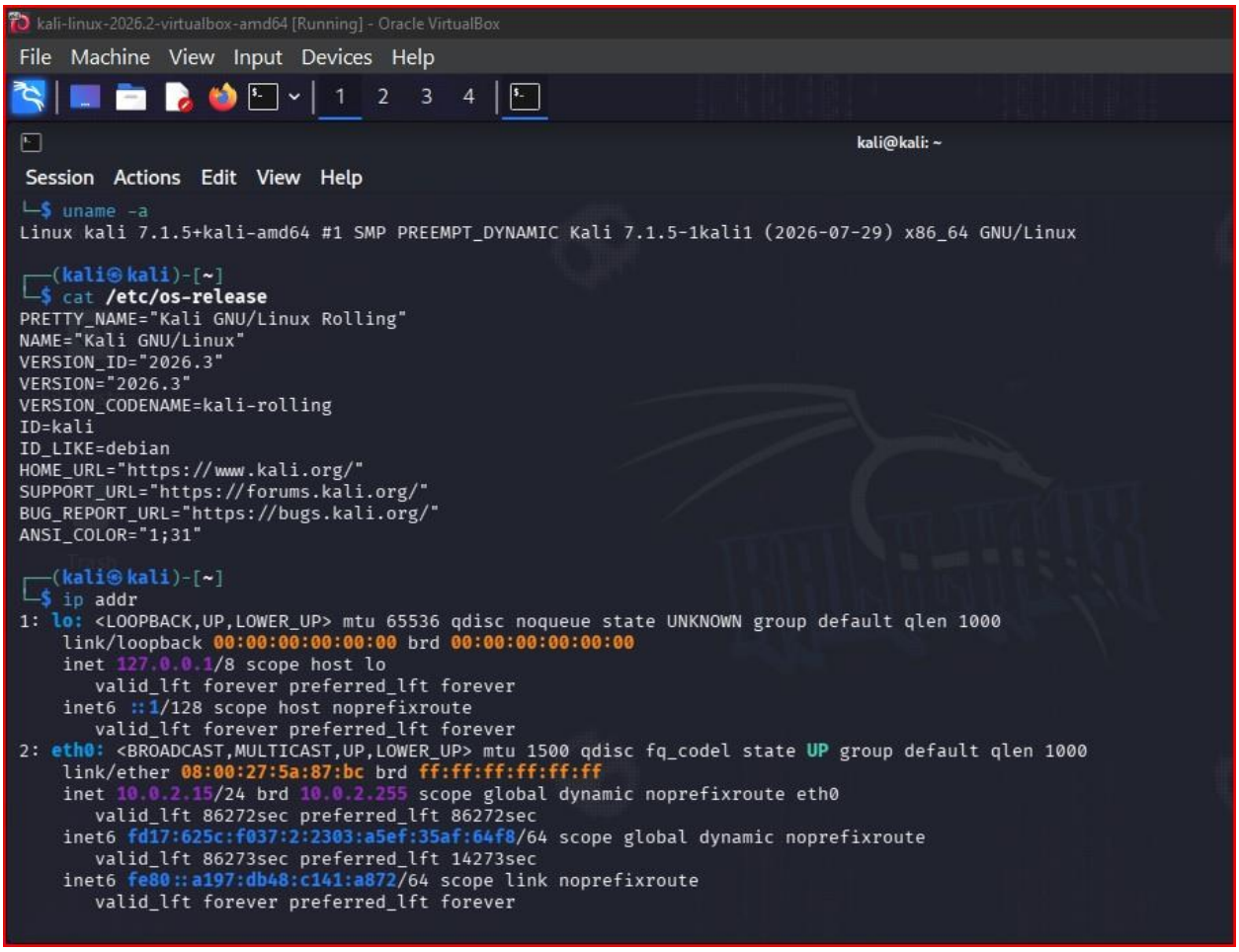
20. Explain the difference between ethical security testing and unauthorized access.

Ethical security testing is performed with proper permission to identify security weaknesses and help improve system protection.

Unauthorized access means accessing or testing someone else's computer, network, or data without permission. It can violate laws and cause harm.

The main difference is that ethical security testing is legal and authorized, while unauthorized access is illegal and unethical.


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ whoami
kali
(kali㉿kali)-[~]
$ hostname
kali
(kali㉿kali)-[~]
$ uname -a
Linux kali 7.1.5+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.1.5-1kali1 (2026-07-29) x86_64 GNU/Linux
(kali㉿kali)-[~]
$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.3"
VERSION="2026.3"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"


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HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
┌──(kali㉿kali)-[~]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 86272sec preferred_lft 86272sec
inet6 fd17:625c:f037:2:2303:a5ef:35af:64f8/64 scope global dynamic noprefixroute
valid_lft 86273sec preferred_lft 14273sec
inet6 fe80::a197:db48:c141:a872/64 scope link noprefixroute
valid_lft forever preferred_lft forever